



9/1/2026

N + - Network Fundamentals & Building Networks.

Network Troubleshooting Tools.

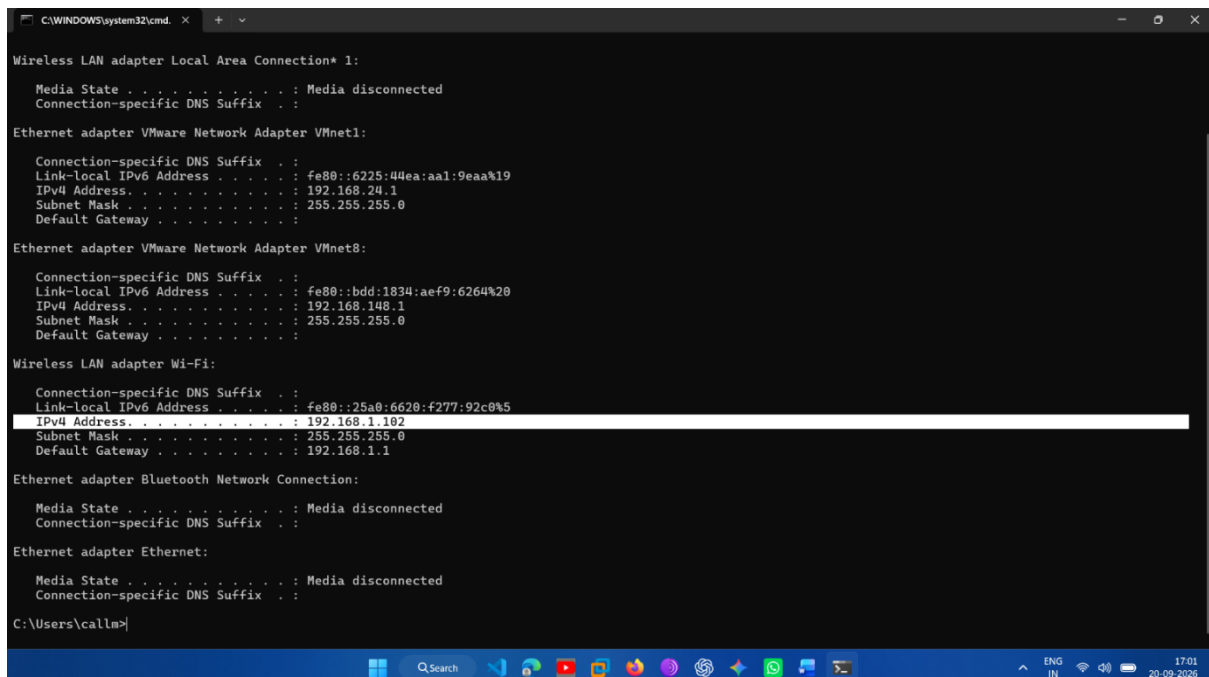
TASK 1.

- ❖ Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

- Find IPv4 Address and Default Gateway:

Command Used:

Ipconfig

A screenshot of a Windows Command Prompt window. The title bar shows the path 'C:\WINDOWS\system32\cmd.' and standard window controls. The command prompt displays the output of the 'ipconfig' command. It lists network adapters: 'Wireless LAN adapter Local Area Connection* 1:' (Media disconnected), 'Ethernet adapter VMware Network Adapter VMnet1:' (IPv4: 192.168.24.1), 'Ethernet adapter VMware Network Adapter VMnet8:' (IPv4: 192.168.148.1), 'Wireless LAN adapter Wi-Fi:' (IPv4: 192.168.1.102, Default Gateway: 192.168.1.1), 'Ethernet adapter Bluetooth Network Connection:' (Media disconnected), and 'Ethernet adapter Ethernet:' (Media disconnected). The 'Wi-Fi' section is highlighted with a white background. The prompt at the bottom is 'C:\Users\callm>'. The Windows taskbar is visible at the bottom with various icons and a system clock showing 17:01 on 20-09-2020.

```
C:\WINDOWS\system32\cmd. x + v

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::6225:40ea:aal:9eaa%19
    IPv4 Address. . . . . : 192.168.24.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::bdd:1834:aef9:6264%28
    IPv4 Address. . . . . : 192.168.148.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::25a0:6620:f277:92c0%5
    IPv4 Address. . . . . : 192.168.1.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\callm>
```

Figure : ipconfig Output.

Result

From the **Wi-Fi adapter**, I found:

- ≈ **IPv4 Address:** 192.168.1.102
- ≈ **Default Gateway:** 192.168.1.1

TASK 2.

- ❖ Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

- Check Connectivity Using Ping:

Command Used:

```
ping www.google.com
```

Result

The ping replies were received successfully, confirming that my computer is connected to www.google.com.

Average Response Time: 9ms

TASK 3.

- ❖ Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

- Trace Route Using Tracert:

Command Used:

```
tracert www.flipkart.com
```

Result

The tracert command successfully traced the route from my computer to www.flipkart.com.

Number of Hops: 9 hops

TASK 4.

❖ Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

➤ Using NSLookup:

Command Used:

```
nslookup www.spotify.com
```

Result

IP Address: 192.168.1.1

Explanation: nslookup helps verify the IP address associated with a website's domain name.

TASK 5.

❖ A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check, Think about checking both network connectivity and DNS resolution.

➤ Troubleshooting Zomato Website Access:

Step	Command	What it Checks
1	ipconfig	Checks the laptop's IP address and default gateway.
2	ping www.zomato.com	Checks whether the laptop has network connectivity to Zomato.
3	nslookup www.zomato.com	Checks whether DNS is correctly resolving Zomato's domain name to an IP address.